KAIZEN MATHS

Lesson starter

Limits and First Principles

First Principles

Name _____

S1

Differentiate each linear function from first principles.

Use first principles to differentiate $f(x) = -3x - 5$.

S2

Differentiate each quadratic function from first principles.

Use first principles to differentiate $f(x) = -x^2 - 5x - 5$.

S3

Differentiate each cubic function from first principles.

Use first principles to differentiate $f(x) = -x^3 + 5x - 2$.

S4

Find each gradient at a point using first principles.

Use first principles to find the gradient of $f(x) = -x^2 - 3$ at $x = 3$.

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ALG-911D-L3 · S

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KAIZEN MATHS

Exit ticket

Limits and First Principles

First Principles

Name _____

EA1

Differentiate each linear function from first principles.

Use first principles to differentiate $f(x) = -5x - 3$.

EA2

Differentiate each quadratic function from first principles.

Use first principles to differentiate $f(x) = -2x^2 + 2x + 2$.

Show your working

KAIZEN MATHS

Exit ticket

Limits and First Principles

First Principles

Name _____

EB1

Differentiate each cubic function from first principles.

Use first principles to differentiate $f(x) = x^3 - 2x + 5$.

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KAIZEN MATHS

Exit ticket

Limits and First Principles

First Principles

Name _____

EA1

Differentiate each linear function from first principles.

Use first principles to differentiate $f(x) = -5x - 3$.

EA2

Differentiate each quadratic function from first principles.

Use first principles to differentiate $f(x) = -2x^2 + 2x + 2$.

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KAIZEN MATHS

Exit ticket

Limits and First Principles

First Principles

Name _____

EB1

Differentiate each cubic function from first principles.

Use first principles to differentiate $f(x) = x^3 - 2x + 5$.

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Limits and First Principles · First Principles

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S1

Differentiate each linear function from first principles.

Use first principles to differentiate $f(x) = -3x - 5$.

Answer:

$f'(x) = -3$

Working:

Start with the first-principles definition.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$\begin{aligned} f(x+h) &= -3(x+h) - 5 \\ &= -3x - 3h - 5 \end{aligned}$$

$$f(x+h) - f(x) = (-3x - 3h - 5) - (-3x - 5)$$

$$= -3h$$

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{-3h}{h} \\ &= \lim_{h \to 0} -3 \\ &= -3 \end{aligned}$$

S2

Differentiate each quadratic function from first principles.

Use first principles to differentiate $f(x) = -x^2 - 5x - 5$.

Answer:

$f'(x) = -2x - 5$

Working:

Start with the first-principles definition.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$f(x+h) = -(x+h)^2 - 5(x+h) - 5$$

$$\begin{aligned} &= -(x^2 + 2xh + h^2) - 5x - 5h - 5 \\ &= -x^2 - 5x - 5 - 2xh - h^2 - 5h \end{aligned}$$

$$\begin{aligned} f(x+h) - f(x) &= (-x^2 - 5x - 5 - 2xh - h^2 - 5h) - (-x^2 - 5x - 5) \\ &= -2xh - h^2 - 5h \\ &= h(-2x - h - 5) \end{aligned}$$

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{h(-2x - h - 5)}{h} \\ &= \lim_{h \to 0} (-2x - h - 5) \\ &= -2x - 5 \end{aligned}$$

Teacher answers & working

Limits and First Principles · First Principles

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S3

Differentiate each cubic function from first principles.

Use first principles to differentiate $f(x) = -x^3 + 5x - 2$.

Answer:

$f'(x) = -3x^2 + 5$

Working:

Start with the first-principles definition.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Use the expansion $(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$.

$$\begin{aligned} f(x+h) &= -(x+h)^3 + 5(x+h) - 2 \\ &= -x^3 + 5x - 2 - 3x^2h - 3xh^2 - h^3 + 5h \\ f(x+h) - f(x) &= (-x^3 + 5x - 2 - 3x^2h - 3xh^2 - h^3 + 5h) - (-x^3 + 5x - 2) \\ &= -3x^2h - 3xh^2 - h^3 + 5h \\ &= h(-3x^2 - 3xh - h^2 + 5) \end{aligned}$$

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{h(-3x^2 - 3xh - h^2 + 5)}{h} \\ &= \lim_{h \to 0} (-3x^2 - 3xh - h^2 + 5) \\ &= -3x^2 + 5 \end{aligned}$$

S4

Find each gradient at a point using first principles.

Use first principles to find the gradient of $f(x) = -x^2 - 3$ at $x = 3$.

Answer:

-6

Working:

Use the point form of the first-principles definition.

$$f'(3) = \lim_{h \to 0} \frac{f(3+h) - f(3)}{h}$$

$$\begin{aligned} f(3+h) &= -(3+h)^2 - 3 \\ &= -(9 + 6h + h^2) - 3 \\ &= -12 - 6h - h^2 \\ f'(3) &= \lim_{h \to 0} \frac{(-12 - 6h - h^2) - -12}{h} \\ f(3) &= -12 \\ &= \lim_{h \to 0} \frac{h(-6 - h)}{h} \\ &= \lim_{h \to 0} (-6 - h) \\ &= -6 \end{aligned}$$

Teacher answers & working

Limits and First Principles · First Principles

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EA1

Differentiate each linear function from first principles.

Use first principles to differentiate $f(x) = -5x - 3$.

Answer:

$f'(x) = -5$

Working:

Start with the first-principles definition.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$\begin{aligned} f(x+h) &= -5(x+h) - 3 \\ &= -5x - 5h - 3 \end{aligned}$$

$$f(x+h) - f(x) = (-5x - 5h - 3) - (-5x - 3) = -5h$$

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{-5h}{h} \\ &= \lim_{h \to 0} -5 \\ &= -5 \end{aligned}$$

EA2

Differentiate each quadratic function from first principles.

Use first principles to differentiate $f(x) = -2x^2 + 2x + 2$.

Answer:

$f'(x) = -4x + 2$

Working:

Start with the first-principles definition.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$\begin{aligned} f(x+h) &= -2(x+h)^2 + 2(x+h) + 2 \\ &= -2(x^2 + 2xh + h^2) + 2x + 2h + 2 \\ &= -2x^2 + 2x + 2 - 4xh - 2h^2 + 2h \end{aligned}$$

$$\begin{aligned} f(x+h) - f(x) &= (-2x^2 + 2x + 2 - 4xh - 2h^2 + 2h) - (-2x^2 + 2x + 2) \\ &= -4xh - 2h^2 + 2h \\ &= h(-4x - 2h + 2) \end{aligned}$$

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{h(-4x - 2h + 2)}{h} \\ &= \lim_{h \to 0} (-4x - 2h + 2) \\ &= -4x + 2 \end{aligned}$$

Teacher answers & working

Limits and First Principles · First Principles

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EB1

Differentiate each cubic function from first principles.

Use first principles to differentiate $f(x) = x^3 - 2x + 5$.

Answer:

$f'(x) = 3x^2 - 2$

Working:

Start with the first-principles definition.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Use the expansion $(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$.

$$\begin{aligned} f(x+h) &= (x+h)^3 - 2(x+h) + 5 \\ &= x^3 + 3x^2h + 3xh^2 + h^3 - 2x - 2h + 5 \\ f(x+h) - f(x) &= (x^3 + 3x^2h + 3xh^2 + h^3 - 2x - 2h + 5) - (x^3 - 2x + 5) \\ &= 3x^2h + 3xh^2 + h^3 - 2h \\ &= h(3x^2 + 3xh + h^2 - 2) \end{aligned}$$

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{h(3x^2 + 3xh + h^2 - 2)}{h} \\ &= \lim_{h \to 0} (3x^2 + 3xh + h^2 - 2) \\ &= 3x^2 - 2 \end{aligned}$$